



DIL Research Assistant Report

To: Policymakers of Chicago

From: Pedro Huet

Date: 09/21/2023

RE: School subsidy randomized control trial analysis and impact evaluation report.

Executive Summary

The literature has argued that subsidies and transfer programs could be effective to increase education and development in communities, so it makes sense to perform experiments to verify the accuracy and magnitude of said variable. An RTC, which employed pairwise randomization treatment assignment, of a proposed subsidy program was conducted in 165 schools across a sample of almost 10,000 elementary-level students located in 2 districts from 2010 to 2012. Two follow-up surveys were conducted after the start of the intervention, one the year after the end of the program, and the second 2 years later. Performing a series of logistic regression models, the data supports the hypothesis that school subsidies reduce the likelihood of school evasion, child pregnancies and marriages during the length of the program, as well as some years afterwards; although the magnitude of the effects are comparatively small when considering other variables. Further, the results of a simple panel difference-in-difference model favors the argument that the subsidy has both short-term and long-term effects on reducing school dropouts. However, the effect of school subsidies wanes when considering that child pregnancies and marriages could also be causes of school dropouts. The conclusion is that school subsidies seem to have a small effect in reducing school evasion, yet the feasibility of this approach isn't as convincing when considering there could be alternative causes to school dropouts which could be employed at a lower cost.

Background & Research

The bolstering of education as a fundamental component of social development has been a topic of great relevance in the last 3 decades. It has also been supported by the literature and multiple studies (Speil et al., 2018; Öztürk et al., 2022). Given this situation, it follows that there could be some policy alternatives to encourage or aid youths' access to this resource. Among the different paths that have been argued to bolster education among communities, many authors have argued in favor of utilizing subsidies, conditional cash/in-kind transfers (Ullmann et al., 2021;). The argument is that giving children and their families access to additional resources for pursuing their studies adds an additional incentives for the recipients to continue pursuing their education.

This argument is enticing, yet the policy it supports seems to implicate the use of a large amount of resources (Loser et al., 2021), meaning it could have an important toll on public finances. In order to assess if subsidies and cash or in-kind transfers are worth the investment, it would make



sense to study the magnitude and reliability of their impact. As a contribution to this task, this report proposes a statistical analysis based on an RTC designed experiment based on a subsidies program targeted towards the education of students.

Study Methodology

The study consist of evaluating an education program that subsidizes the cost of education for students through in-kind transfers. The study consisted of a program that lasted for three years, from 2010 to 2012, and it targeted children enrolled in public schools. Throughout this time period, the subsidies were delivered at the beginning of every school year to the cohort of students enrolled in grade 6 at the start of the program.

To evaluate its effectiveness, an experiment was designed using pairwise randomization to assign schools within two districts to a treatment or control group. The main unit of analysis are the children whose schools did or did not receive the subsidies. All children enrolled in the treatment schools received the subsidy for the three years, conditional that they were still enrolled in school. To conduct the study, baseline data was collected for the schools and cohorts, and two follow-up surveys were conducted: the first one, on the year after the last subsidy was offered to students (year 3), and another two years later (year 5).

With regards to performance metrics for the program, the study considered school evasion, if a student dropped out of his or her educational formation, as the primary outcome of interest. It considered teen pregnancy and teen marriage as secondary variables of interest.

Based on the available data, it would seem reasonable to choose a series of variables that likely have a larger effect on affecting school evasion. The first and most relevant variable for the study is treatment (dichotomous variable), if the student was in a school that gave him or her access to the subsidy, as this result allows us to approach the effectiveness of subsidies and transfer programs. A second variable would be the sex of the student, as its possible that male or female students, due to social and contextual factors, may have a larger tendency to dropout of their education. A third variable is the year of birth of students, since younger students tend to have less motivation of dropping out of school (they are usually less qualified at finding alternative venues of employment than older students). Since this variable is categorized by the year of birth of the students receiving the subsidies (between 1985 and 1992), the variable can be treated as ordinal (continuous, for our purposes). Finally, we also consider the month in which the visit was made to students (nominal variable), since months closer to the start and end of the school year (June, July, August, and December) are likely to be periods in which the students are more prone to end their studying experiencing (students could be more likely to want to finish their semesters due to a process of inertia).

With regards to the dependent variables, school dropouts (if the student left school), teen pregnancies (if a student or his partner experienced a pregnancy) and marriages (if the student got



married) are measured as dichotomous variables (from 0 to 1). These variables will be measured both during the year 3 since the start of the intervention and on year 5 of it occurring.

The study consists of a series of a maximum likelihood logarithmic models estimating the short-term and long-term effects of the intervention, as well as a difference-in-difference panel linear model. The advantage of using a non-linear model is that it is measuring the probabilities of an outcome occurring, in this case, a student dropping out, rather than a continuous quantity (i.e. “the degree in which a student drops out”). Because of this, a non-linear model like a logistic model is more versatile than a regular linear probability model, meaning it’ll likely make a more useful estimation. The main model can be summarized by the following equation.

$$\text{School Dropouts} = B_0 + B_1(\text{treatment}) + B_2(\text{sex}) + B_3(\text{year of birth}) + B_4(\text{month of visit}) + u$$

With regards to the linear fixed effects difference-in-difference model, since the panel data only consists of two years, it doesn’t make sense to use more sophisticated models (i.e. Difference-in-Difference-in-Differences model or Multiple-Timming-Fixed Effects Models), as there isn’t enough data to obtain noticeably different estimations. Perhaps the main problem on the outset of this study is that we cannot verify the Parallel Trends Assumption: it isn’t clear that the individuals who were treated on the third year of the program followed the same trend as those who were untreated during this period. Although it would have been preferable to perform this analysis just before the start of the intervention to have more certainty of both groups being comparable, an analysis of this sort can still be insightful to observe if there were differences between the effects of the subsidies at the 3rd year of the intervention and the 5th year. The panel model can be summarized by the following equation.

$$\text{School Dropouts} = B_0 + B_1(\text{treatment}) + B_2(\text{long-term}) + B_3(\text{treatment} * \text{long-term}) + u$$

Key Findings

Before starting the analysis, it is important to make sure that the Randomized Control Trial worked in the correct manner. Table 1 shows that a multivariate regression model that includes the variables that could have been affecting the school’s assignment does not find much evidence of bias in the assignment. Aside from the number of latrines the schools have, none of the other variables related to the schools (number of teachers and students, average student scores, rural location, etc.) seem to be associated to a significance level of above 10% with being assigned to the treatment group; while none of the variables reaches a confidence level of above 5%. Because of this, it seems that, in general, the baseline characteristics seem balanced.

Table 1

Logistic regression

Number of obs = 155

LR chi2(11) = 11.61

Prob > chi2 = 0.3935

Pseudo R2 = 0.0541

Log likelihood = -101.60301

treatment	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
district	-.2465828	.4286512	-0.58	0.565	-1.086724	.5935582
location	-.8957887	.6688227	-1.34	0.180	-2.206657	.4150797
n_teachers	-.0612822	.1076023	-0.57	0.569	-.2721788	.1496143
n_teachers_fem	.0043786	.0850735	0.05	0.959	-.1623624	.1711195
female_head_teacher	.298549	.611635	0.49	0.625	-.9002335	1.497332
n_students_fem	-.0056233	.005323	-1.06	0.291	-.0160562	.0048097
n_students_male	.0016141	.0054949	0.29	0.769	-.0091557	.0123839
n_schools_2km	-.040937	.1052616	-0.39	0.697	-.2472459	.165372
av_teacher_age	.0807291	.0601273	1.34	0.179	-.0371183	.1985764
av_student_score	.0057991	.0070456	0.82	0.410	-.0080101	.0196083
n_latrines	.074235	.038603	1.92	0.054	-.0014256	.1498955
_cons	-2.694787	3.411237	-0.79	0.430	-9.38069	3.991115

With regards to the short-term effects of the subsidies, Tables 2, 3 and 4 show the possible short-term effects of receiving subsidies on children's education. Table 2 shows that treatment into the group after 3 years has a negative and statistically significant effect, at a significance level of above 1%, in the student dropping out. When seeing the magnitude, its size is around 2.2%, meaning that, if all other variables remain equal, a student that is in a school that receives subsidies is 2.2% less likely to drop-out of school. A student's year of birth also show statistically significant negative effects on school evasion, meaning that, for each year a student is younger, all else equal, he or she is almost 6% less likely to dropout. When observing sex, we observe that being a female student, all else equal, seems to be associated with a 8.6% probability of being more prone to dropout of school. Finally, the students who were surveyed in October were less prone to dropout by around 9%, all else equal. The results of these variables are all significant at a level of above 5%. Figure 1 offers a visual summary of the magnitude and significance level of the models' findings.



Table 2

Logistic regression Number of obs = 9,004
LR chi2(7) = 787.01
Prob > chi2 = 0.0000
 Log likelihood = -3222.5966 Pseudo R2 = 0.1088

dropout	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
trt_school	-.2130132	.0649831	-3.28	0.001	-.3403776	-.0856488
yob	-.5882476	.0231562	-25.40	0.000	-.633633	-.5428623
sex	.8022253	.0684522	11.72	0.000	.6680614	.9363893
visit_month						
9	-.4789905	.2690345	-1.78	0.075	-1.006288	.0483074
10	-.8116111	.2812518	-2.89	0.004	-1.362854	-.2603678
11	-.1588488	.2698362	-0.59	0.556	-.6877179	.3700204
12	-.1925838	.3149806	-0.61	0.541	-.8099344	.4247669
_cons	1168.343	46.05379	25.37	0.000	1078.079	1258.607

Average marginal effects Number of obs = 9,004
 Model VCE: OIM

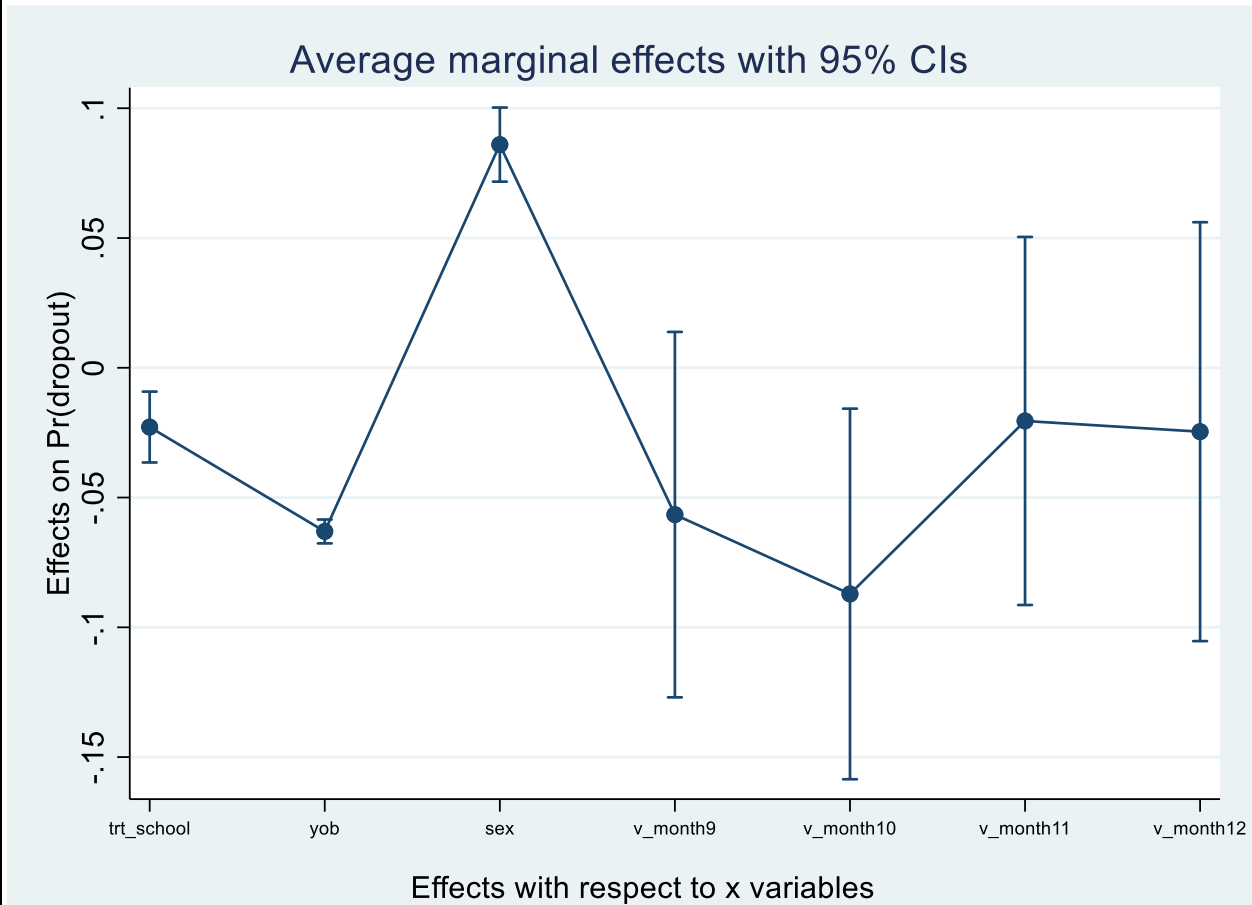
Expression: Pr(dropout), predict()
 dy/dx wrt: trt_school yob sex 9.visit_month 10.visit_month 11.visit_month 12.visit_month

	Delta-method					
	dy/dx	std. err.	z	P> z	[95% conf. interval]	
trt_school	-.022835	.0069611	-3.28	0.001	-.0364786	-.0091915
yob	-.0630602	.0023503	-26.83	0.000	-.0676666	-.0584538
sex	.0859986	.0072841	11.81	0.000	.071722	.1002753
visit_month						
9	-.0565799	.0359211	-1.58	0.115	-.1269839	.0138242
10	-.0871516	.0364265	-2.39	0.017	-.1585462	-.015757
11	-.0204904	.0361792	-0.57	0.571	-.0914004	.0504196
12	-.0246181	.0411715	-0.60	0.550	-.1053128	.0560767

Note: dy/dx for factor levels is the discrete change from the base level.



Figure. 1



Tables 3 and 4 offer similar results. They show that, as with school evasion, it would also seem that the subsidies program has negative and statistically significant effects (of a significance level of above 95%) towards experiencing teen pregnancy and marriages, yet it also seems relatively small (around 2% reduction).



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Table 3

Logistic regression

Number of obs = 4,328

LR chi2(6) = 465.06

Prob > chi2 = 0.0000

Log likelihood = -1328.7058

Pseudo R2 = 0.1489

	pregnt	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
trt_school		-.2561749	.1026331	-2.50	0.013	-.4573321	-.0550178
yob		-.7762106	.040134	-19.34	0.000	-.8548718	-.6975493
sex		0	(omitted)				
visit_month							
9		.2279757	.530794	0.43	0.668	-.8123613	1.268313
10		-.3226972	.5488073	-0.59	0.557	-1.39834	.7529454
11		.5934452	.5323371	1.11	0.265	-.4499164	1.636807
12		.1519605	.614408	0.25	0.805	-1.052257	1.356178
_cons		1541.963	79.8308	19.32	0.000	1385.498	1698.429

Average marginal effects

Number of obs = 4,328

Model VCE: OIM

Expression: Pr(pregnt), predict()

dy/dx wrt: trt_school yob sex 9.visit_month 10.visit_month 11.visit_month 12.visit_month

		Delta-method				[95% conf. interval]	
		dy/dx	std. err.	z	P> z		
trt_school		-.0228837	.0091636	-2.50	0.013	-.040844	-.0049233
yob		-.0693375	.0034071	-20.35	0.000	-.0760153	-.0626597
sex		0	(omitted)				
visit_month							
9		.0184215	.0398947	0.46	0.644	-.0597706	.0966136
10		-.0216869	.0404933	-0.54	0.592	-.1010523	.0576784
11		.0539865	.0403937	1.34	0.181	-.0251837	.1331566
12		.0119742	.0473682	0.25	0.800	-.0808657	.104814

Note: dy/dx for factor levels is the discrete change from the base level.



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Table 4

Logistic regression

Log likelihood = -1414.583

Number of obs = 8,791
LR chi2(7) = 774.77
Prob > chi2 = 0.0000
Pseudo R2 = 0.2150

married	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
trt_school	-.3384766	.1045114	-3.24	0.001	-.5433152	-.133638
yob	-.7737201	.0377909	-20.47	0.000	-.8477889	-.6996513
sex	2.625267	.1480263	17.74	0.000	2.335141	2.915394
visit_month						
9	.0700375	.5211146	0.13	0.893	-.9513283	1.091403
10	-.700827	.5449954	-1.29	0.198	-1.768998	.3673443
11	.3370815	.5230385	0.64	0.519	-.6880551	1.362218
12	.2403885	.5833123	0.41	0.680	-.9028827	1.38366
_cons	1534.292	75.1138	20.43	0.000	1387.072	1681.513

. margins, dydx(*)

Average marginal effects
Model VCE: OIM

Number of obs = 8,791

Expression: Pr(married), predict()

dy/dx wrt: trt_school yob sex 9.visit_month 10.visit_month 11.visit_month 12.visit_month

	Delta-method					
	dy/dx	std. err.	z	P> z	[95% conf. interval]	
trt_school	-.0144753	.0044809	-3.23	0.001	-.0232577	-.0056928
yob	-.0330889	.0017358	-19.06	0.000	-.036491	-.0296868
sex	.1122722	.0068607	16.36	0.000	.0988254	.125719
visit_month						
9	.0029293	.021273	0.14	0.890	-.038765	.0446236
10	-.0222196	.0214501	-1.04	0.300	-.0642611	.0198218
11	.0155237	.0215015	0.72	0.470	-.0266184	.0576658
12	.0106921	.0248125	0.43	0.667	-.0379395	.0593236

Note: dy/dx for factor levels is the discrete change from the base level.



With regards to long-term effects, Tables 5, 6 and 7 show that the results remained similar: if a student receives subsidies, his or her dropout rate decreased by around 5%, which is slightly higher than what was found in the short term? at a level of statistical significance of above 5%. Perhaps the most important difference in the long-term can be seen in Table 7: the negative effect of subsidies on teen marriage seems to decrease noticeably after 5 years of the program's start, which is reflected in the fact that the significance level of this variable decrease to below 5% , but comfortably above 10%, and the magnitude of the effect decreased to around 1%.

Table 5

Logistic regression Number of obs = 8,990
LR chi2(5) = 858.02
Prob > chi2 = 0.0000
 Log likelihood = -4538.2994 Pseudo R2 = 0.0864

dropout	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
trt_school	-.3251957	.0526112	-6.18	0.000	-.4283118	-.2220795
yob	-.4891332	.0190829	-25.63	0.000	-.5265351	-.4517313
sex	.7239195	.0546015	13.26	0.000	.6169025	.8309366
visit_month						
5	.2983619	.0757346	3.94	0.000	.1499247	.4467991
7	.1621009	.0686074	2.36	0.018	.0276329	.2965689
_cons	971.4962	37.95153	25.60	0.000	897.1126	1045.88

Average marginal effects Number of obs = 8,990
 Model VCE: OIM

Expression: Pr(dropout), predict()
 dy/dx wrt: trt_school yob sex 5.visit_month 7.visit_month

	Delta-method					
	dy/dx	std. err.	z	P> z	[95% conf. interval]	
trt_school	-.0537056	.0086365	-6.22	0.000	-.0706327	-.0367784
yob	-.0807796	.002787	-28.98	0.000	-.0862421	-.0753171
sex	.1195542	.008775	13.62	0.000	.1023554	.1367529
visit_month						
5	.0490473	.0123383	3.98	0.000	.0248647	.0732299
7	.0258428	.0107618	2.40	0.016	.0047501	.0469355

Note: dy/dx for factor levels is the discrete change from the base level.



Table 6

Logistic regression

Log likelihood = -2058.9237

Number of obs = 3,974
 LR chi2(4) = 555.27
 Prob > chi2 = 0.0000
 Pseudo R2 = 0.1188

pregnt	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
trt_school	-.2683974	.0775899	-3.46	0.001	-.4204708	-.116324
yob	-.687464	.0327906	-20.97	0.000	-.7517324	-.6231956
sex	0	(omitted)				
visit_month						
5	.0621545	.1149647	0.54	0.589	-.1631722	.2874813
7	.3013065	.1019705	2.95	0.003	.1014479	.5011651
_cons	1366.754	65.23754	20.95	0.000	1238.89	1494.617

Average marginal effects

Model VCE: OIM

Number of obs = 3,974

Expression: Pr(pregnt), predict()
 dy/dx wrt: trt_school yob sex 5.visit_month 7.visit_month

	Delta-method					
	dy/dx	std. err.	z	P> z	[95% conf. interval]	
trt_school	-.0459515	.0132241	-3.47	0.001	-.0718702	-.0200328
yob	-.1176987	.0045147	-26.07	0.000	-.1265474	-.1088499
sex	0	(omitted)				
visit_month						
5	.0100682	.0185936	0.54	0.588	-.0263746	.046511
7	.0511042	.0168327	3.04	0.002	.0181128	.0840956

Note: dy/dx for factor levels is the discrete change from the base level.

Table 7



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Logistic regression
Log likelihood = -2536.9362

Number of obs = 8,063
LR chi2(5) = 1306.63
Prob > chi2 = 0.0000
Pseudo R2 = 0.2048
```

	married	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
trt_school		-.1371421	.0728537	-1.88	0.060	-.2799328	.0056486
yob		-.713078	.0280966	-25.38	0.000	-.7681465	-.6580096
sex		2.423646	.09585	25.29	0.000	2.235783	2.611508
visit_month							
5		-.0470325	.1081712	-0.43	0.664	-.2590442	.1649792
7		.1886655	.0966351	1.95	0.051	-.0007359	.3780669
_cons		1414.974	55.852	25.33	0.000	1305.506	1524.442

```
Average marginal effects
Model VCE: OIM

Number of obs = 8,063
```

```
Expression: Pr(married), predict()
dy/dx wrt: trt_school yob sex 5.visit_month 7.visit_month
```

		Delta-method				[95% conf. interval]	
		dy/dx	std. err.	z	P> z		
trt_school		-.0130801	.0069457	-1.88	0.060	-.0266934	.0005333
yob		-.0680105	.0024052	-28.28	0.000	-.0727246	-.0632965
sex		.2311577	.0084983	27.20	0.000	.2145014	.247814
visit_month							
5		-.0042307	.0097508	-0.43	0.664	-.023342	.0148806
7		.0180713	.0090502	2.00	0.046	.0003332	.0358093

Note: dy/dx for factor levels is the discrete change from the base level.

Differenece in Differenece approach.

Having seen the previous models, it would appear the effect of the subsidies on school dropout exists both in the short term and the long-term, being slightly higher in the long term. However, it would make sense to try to get the differentiated effect of each period. For that, a panel model is employed (see Figure 2). The results can be visualized in Table 8. According to these, it would appear that the subsidies intervention has a negative effect in school evasion in the short-term as well as the long-term, it being statistically significant at a level of above 5%. What this seems to imply is that the total effect of subsidies on school

evasion, when comparado to somewone who didn't receive these is around 2.5% in the short term and around 2.7% less likelihood of dropping out of school.

Table 8

Linear regression				Number of obs	=	19,310	
				F(3, 19306)	=	114.39	
				Prob > F	=	0.0000	
				R-squared	=	0.0177	
				Root MSE	=	.41242	

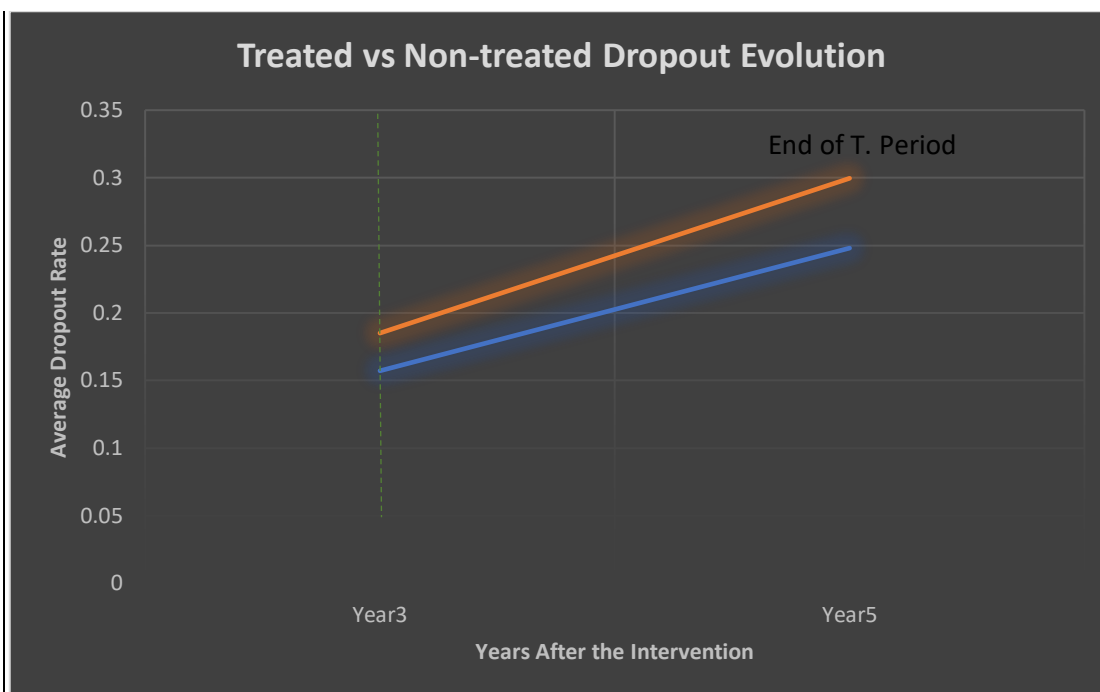
	dropout		Coefficient	Robust std. err.	t	P> t	[95% conf. interval]

	1.trt_school		-.0278977	.0076554	-3.64	0.000	-.0429031 - .0128924
	1.post		.1144972	.0085763	13.35	0.000	.0976869 .1313074

	trt_school#post						
	1 1		-.023748	.0118629	-2.00	0.045	-.0470003 - .0004958

	_cons		.1850947	.0055426	33.40	0.000	.1742308 .1959586

Figure 2



Relationship between school marriage, children and dropouts?

Although it makes sense to assume that dropping out of school and teen marriage and pregnancy shouldn't be included into the regression due to possible bias with dropping out (both could be a product of a third variable, like irresponsibility), it is possible to think that teen marriage and teen pregnancy could be relevant variables to explaining the process of dropping out of school. When estimating a regression with these two variables in Tables 9 and 10, we see that the impact of being treated decrease substantially while the explanatory effect of these variables increase, especially at the short-term. This could mean that more attention should be put into the causality of these variables, and not just assuming they are both perforando metrics.

Table 9

```
logit dropout married children trt_school if post == 0
```

Logistic regression

Number of obs = 4,679

LR chi2(3) = 2315.06

Prob > chi2 = 0.0000

Log likelihood = -987.52283

Pseudo R2 = 0.5396



dropout	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
married	4.327641	.1994044	21.70	0.000	3.936815	4.718466
children	4.23428	.2252383	18.80	0.000	3.792821	4.675739
trt_school	-.1336577	.1269921	-1.05	0.293	-.3825577	.1152423
_cons	-2.825353	.0928272	-30.44	0.000	-3.007291	-2.643415

Table 10

Logistic regression
 Number of obs = 4,231
 LR chi2(3) = 2336.73
 Prob > chi2 = 0.0000
 Log likelihood = -1273.7816
 Pseudo R2 = 0.4784

dropout	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
married	2.109351	.1208567	17.45	0.000	1.872476	2.346226
children	2.685343	.1188434	22.60	0.000	2.452414	2.918271
trt_school	-.2074639	.1057219	-1.96	0.050	-.4146751	-.0002528
_cons	-2.577347	.0866784	-29.73	0.000	-2.747233	-2.40746

. margins, dydx(*)

Average marginal effects
 Model VCE: OIM
 Number of obs = 4,231

Expression: Pr(dropout), predict()
 dy/dx wrt: married children trt_school

		Delta-method				[95% conf. interval]	
	dy/dx	std. err.	z	P> z			
married	.1788809	.0096967	18.45	0.000	.1598757	.197886	
children	.2277271	.0086478	26.33	0.000	.2107777	.2446765	
trt_school	-.0175937	.0089682	-1.96	0.050	-.035171	-.0000164	

Conclusions and Recommendations

The models estimating the effects of the intervention show that subsidies and transfer programs seem to have a negative and statistically significant effect on school evasion, teen pregnancies, and marriages. However, when observing the magnitude, its size is relatively small in both the short and long-term. Further, when estimating a regression that includes teen marriage and children as explanatory variables, the impact of being treated into the program decreases substantially. This seems to imply that dropping out of school is heavily influenced by these variables: the subsidy's direct effect on educational formation is not as convincing as initially argued by the literature. Because of this, it follows that the feasibility of this policy alternative isn't as clear when considering the large amount of resources it requires and the toll it could have on public finances. It would make sense for policymakers to put more attention into the causality of teen pregnancies and marriages, rather than just assuming subsidies alone will fix the dropping out process. Therefore, it is recommended to explore alternative programs that address these specific causes to school dropouts, which could likely be employed at a lower cost.

Appendix

Table A1

c) Possible Interaction Effects of the Subsidy (Mentioned, but placed in Appendix)

Iteration 0: log likelihood = -3616.1031
 Iteration 1: log likelihood = -3256.3472
 Iteration 2: log likelihood = -3222.3153
 Iteration 3: log likelihood = -3222.2188
 Iteration 4: log likelihood = -3222.2188

Logistic regression

Number of obs = 9,004
 LR chi2(8) = 787.77
 Prob > chi2 = 0.0000
 Pseudo R2 = 0.1089

Log likelihood = -3222.2188

dropout	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
1.sex	.8553479	.0919104	9.31	0.000	.6752069	1.035489
1.trt_school	-.1497789	.0974509	-1.54	0.124	-.3407792	.0412214
sex#trt_school						
1 1	-.1130723	.1300684	-0.87	0.385	-.3680017	.1418571
yob	-.588758	.0231685	-25.41	0.000	-.6341675	-.5433485
visit_month						
9	-.4786788	.269078	-1.78	0.075	-1.006062	.0487045
10	-.8135247	.2813077	-2.89	0.004	-1.364878	-.2621717
11	-.1599604	.2698813	-0.59	0.553	-.6889181	.3689972
12	-.1935362	.3150058	-0.61	0.539	-.8109362	.4238637
_cons	1169.329	46.07736	25.38	0.000	1079.019	1259.639

.



Table A2

. logit dropout yob##trt_school sex i.visit_month if post == 0

Iteration 0: log likelihood = -3616.1031
 Iteration 1: log likelihood = -3287.2836
 Iteration 2: log likelihood = -3225.7565
 Iteration 3: log likelihood = -3214.0638
 Iteration 4: log likelihood = -3213.9853
 Iteration 5: log likelihood = -3213.9852

Logistic regression

Log likelihood = -3213.9852

Number of obs = 9,004
 LR chi2(20) = 804.24
 Prob > chi2 = 0.0000
 Pseudo R2 = 0.1112

dropout	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
yob						
1986	-.6087783	.3355975	-1.81	0.070	-1.266537	.0489807
1987	-1.342022	.2951157	-4.55	0.000	-1.920438	-.7636056
1988	-1.846885	.2800821	-6.59	0.000	-2.395836	-1.297934
1989	-2.488037	.2780185	-8.95	0.000	-3.032943	-1.943131
1990	-3.067124	.2808695	-10.92	0.000	-3.617619	-2.51663
1991	-3.708385	.3010533	-12.32	0.000	-4.298439	-3.118332
1992	-3.654042	.362087	-10.09	0.000	-4.363719	-2.944364
1.trt_school	-.3982257	.4187955	-0.95	0.342	-1.21905	.4225984
yob#trt_school						
1986 1	.0298708	.5236579	0.06	0.955	-.9964797	1.056221
1987 1	.1919541	.4610569	0.42	0.677	-.7117008	1.095609
1988 1	.2949508	.4407094	0.67	0.503	-.5688237	1.158725
1989 1	.2607892	.436644	0.60	0.550	-.5950174	1.116596
1990 1	.028782	.442476	0.07	0.948	-.838455	.896019
1991 1	.10818	.4708202	0.23	0.818	-.8146107	1.030971
1992 1	.3134019	.5543103	0.57	0.572	-.7730263	1.39983
sex	.8105168	.0688728	11.77	0.000	.6755287	.945505
visit_month						
9	-.4962747	.2703959	-1.84	0.066	-1.026241	.0336915
10	-.8174601	.2824944	-2.89	0.004	-1.371139	-.2637813
11	-.1663211	.271211	-0.61	0.540	-.6978849	.3652427
12	-.2035364	.3163445	-0.64	0.520	-.8235603	.4164875
_cons	.784527	.3731042	2.10	0.035	.0532563	1.515798



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